Use case name: Create a transport
Textual Description: The system gets an items list, a branch and a designated delivery date and time. The system creates transport which consists of transport routes from supplier(s) to branch. We attach a truck and a relevant driver.
List of Actors: Transport Manager, Driver.
Pre-conditions: There exists a Stocker at the branch at the given delivery date-time. There exists a suitable supplier for at least part of the required items. There exists a relevant truck for the delivery at the required time. And there exists a driver for the relevant time frame with a suitable license type.
Post-conditions: We create a transport with association to transport routes which associates to a delivery file, the transport also associates with a truck and a driver.
Main success scenario: INPUT: itemsList, branchId, deliveryDateTime 1. VERIFY there is a Stocker available at the branch at deliveryDateTime 2. FIND suppliers who can supply at least part of the items 3. IF no suitable supplier found → FAIL 4. FIND a truck that is available at deliveryDateTime 5. IF no truck found → FAIL 6. FIND a driver who: - is available at deliveryDateTime - has a valid license for the truck 7. IF no suitable driver found → FAIL 8. CREATE Transport entity 9. FOR each supplier: a. CREATE TransportRoute from supplier to branch b. CREATE DeliveryFile for route c. ASSOCIATE DeliveryFile with TransportRoute d. ADD TransportRoute to Transport 10. ASSOCIATE truck and driver with the Transport 11. SAVE Transport to database OUTPUT: Transport created successfully

Alternatives/Extensions:

IF suppliers.isEmpty():

DISPLAY "No supplier found for required items"

ABORT transport creation

IF truck == null:

DISPLAY "No truck available at the required time"

ABORT transport creation

IF driver == null:

DISPLAY "No driver available or valid for selected truck"

ABORT transport creation

IF only some items can be fulfilled:

CREATE transport routes for available items

FLAG transport as "Partially fulfilled"

IF truck or driver already assigned to another transport at same time:

DISPLAY "Truck or driver already in use"

ALLOW user to select different resources or reschedule

Contract:

Operation: createTransport(itemsList : List<Item>, branchId : int, deliveryDateTime : datetime)

References:
Use Case: Create a Transport

Preconditions:

- There is a Stocker available at the specified branch at deliveryDateTime.
- There exists at least one supplier that can supply part of the requested items.
- There is a truck available for delivery at the required time.
- There is a driver available during the time window with a valid license for the selected truck.

Postconditions:

- A Transport instance was **created**.
- The Transport was **associated** with one or more TransportRoute instances.
- Each TransportRoute was **associated** with a DeliveryFile.
- The Transport was **associated** with a selected Truck.
- The Transport was **associated** with a selected Driver.